Springer Water

Vangala Sunitha · Bandi Muralidhara Reddy ·
Yenugu Sudharshan Reddy ·
Mannala Prasad · Badapalli Pradeep Kumar ·
Etikala Balaji *Editors*

Remote Sensing, GIS and Modelling for Water Resource Management

Volume 2

Springer

Springer Water

The book series Springer Water comprises a broad portfolio of multi- and interdisciplinary scientific books, aiming at researchers, students, and everyone interested in water-related science. The series includes peer-reviewed monographs, edited volumes, textbooks, and conference proceedings. Its volumes combine all kinds of water-related research areas, such as: the movement, distribution and quality of freshwater; water resources; the quality and pollution of water and its influence on health; the water industry including drinking water, wastewater, and desalination services and technologies; water history; as well as water management and the governmental, political, developmental, and ethical aspects of water.

Vangala Sunitha · Bandi Muralidhara Reddy ·
Yenugu Sudharshan Reddy · Mannala Prasad ·
Badapalli Pradeep Kumar · Etikala Balaji
Editors

Remote Sensing, GIS and Modelling for Water Resource Management

Volume 2

Springer

Editors
Vangala Sunitha
Department of Geology
Yogi Vemana University
Kadapa, Andhra Pradesh, India

Yenugu Sudharshan Reddy
Department of Civil Engineering
Sai Rajeswari Institute of Technology
Proddatur, Andhra Pradesh, India

Badapalli Pradeep Kumar
CSIR-National Geophysical Research Institute
Hyderabad, Telangana, India

Bandi Muralidhara Reddy
Department of Earth Sciences
Yogi Vemana University
Kadapa, Andhra Pradesh, India

Mannala Prasad
Department of Geology
Central Tribal University of Andhra Pradesh
Vizianagaram, Andhra Pradesh, India

Etikala Balaji
Yogi Vemana University
Kadapa, Andhra Pradesh, India

ISSN 2364-6934 ISSN 2364-8198 (electronic)
Springer Water
ISBN 978-3-031-99496-8 ISBN 978-3-031-99497-5 (eBook)
https://doi.org/10.1007/978-3-031-99497-5

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

Preface

The current book comprehensively explores cutting-edge technologies for water resource management for ensuring sustainable development. It delves into a range of promising tools like Remote Sensing, GIS, machine learning and hydrological modelling, and Monte-Carlo Simulation all of which are crucial for water resource management to attain sustainable development goals. The content includes detailed reviews, original research, and real-world case studies on Remote Sensing and GIS modelling, Aquifer mapping, soil properties, and groundwater dynamics, MIF and AHP techniques, MCDM approach, fuzzy modelling, LULC impact on water resources, Monte-Carlo Simulation, policies, frameworks for effective water resource management for sustainable development. It will be useful for researchers, policy-makers, urban planners, etc. by offering them an indispensable resource for the latest and most promising technologies in water resource management.

Kadapa, India	Vangala Sunitha
Kadapa, India	Bandi Muralidhara Reddy
Proddatur, India	Yenugu Sudharshan Reddy
Vizianagaram, India	Mannala Prasad
Hyderabad, India	Badapalli Pradeep Kumar
Kadapa, India	Etikala Balaji